MICHAEL KING

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SUMMARY

Mechanical engineer with five years at a robotics manufacturer, where the job grew from production engineering into owning the company's ERP, its software release process, and its AI platform adoption. I turn operations problems into working, tested software, building with AI assistance and verifying everything before it ships. On my own time I build and operate automated data systems and edge hardware.

PROFESSIONAL EXPERIENCE

Plus One Robotics (robotic parcel-handling systems) · San Antonio, TX · Mar 2021 – Present

Operations Systems Engineer II
Mar 2026 – Present

- Sole developer of InductOne Tools, a custom Frappe/ERPNext application that runs our robot product line from engineering configuration through supplier handoff to field service. About 33,500 lines, 50 custom DocTypes, 40 server-side API methods, and 24 migration patches, built over 5.5 months. The system now carries roughly 1,100 items, 277 BOMs, 93 engineering signoffs, and 90 allocated part numbers for a 30-user organization.
- Replaced a manual builder-package process that took an estimated 200 hours per machine. The same generator now produces RFQ packages too. Modeled value is $43K to $56K per year of recovered engineering time at our current build rate.
- Integrated ERPNext with GitLab, the production imaging system, and device log imports from IPC, HMI, and camera units. Use AI agents with access to GitLab and Microsoft 365 in daily work.
- Specified and directed the build of a desk-side hardware test rig: two IPCs, an HMI, a PLC, and two PoE cameras networked as a complete InductOne cell, so software and communications are exercised on real hardware before release.
- Replaced spreadsheet part-number allocation with an ERPNext batch-allocation process; every new part now records its initial allocation and links to its Engineering Change ticket in GitLab.
- Alongside the systems work: build every new production IPC myself, run inventory including the physical counts for the company-wide audit, and manage freight and parcel shipping accounts end to end.
- Head Ops assembly of the Individual Cup Control gripper. Turned an ambiguous, unsupported process into an in-field maintenance test and compensator replacement guide, shifting routine service to the customer with Product and Engineering approval.
- Built the external-builder workflow. Suppliers see only their own builds through row-level permissions, releases pass through readiness gates, and acceptance creates a locked as-built record in a single transaction.
- Lead the company's AI platform evaluation and use multi-agent workflows on real work every day. I review agent output for PII, data provenance, and untested claims before anything ships.
- Hold the Operations sign-off on product software releases. Our product ships as orchestrated Docker images assembled into deployment packages, and approval requires linked evidence for every requirement: build artifacts, documentation merge requests, QA results, and executive approvals.

Operations Systems Engineer
Nov 2025 – Mar 2026

- Wrote the company's configuration status accounting framework and serialization strategy (ISO 10007 and EIA-649C). Designed the BOM architecture and variant logic for high-mix, low-volume manufacturing.
- Worked with the VP of Operations and engineering leadership to find and close gaps between departments.

Production Engineer
Dec 2022 – Dec 2025

- Owned the production systems between engineering and manufacturing. Built commissioning tooling and runbooks for robot deployments in the US and EU.
- Set up and ran the OS imaging infrastructure for our Ubuntu production machines, including LAN-based deployment and remote troubleshooting of live units.
- Built shop-floor tools, including an OpenCV camera station that verifies QA labels with OCR, packaged as a Windows executable so operators could just run it.

Systems Integration Engineering Intern
Mar 2021 – Dec 2022

- Wrote Python GUI and CLI tools that error-proofed documentation and manufacturing QA records. Supported hardware deployment, then converted to full-time.

INDEPENDENT PROJECTS

- **OnScript** – Collects every U.S. congressional press release daily, measures shared party language, and publishes a static site where each claim links to its sources. Runs unattended on GitHub Actions with fail-closed checks, a published methodology, and a corrections log. onscript.news · github.com/mlawsonking/onscript
- **The Exhaust** – Automated collectors over public NHTSA, CMS, WARN, and CPSC data, with preregistered evaluations. The first two forecasts missed their targets and the writeups are published as they happened. github.com/mlawsonking/theexhaust
- **Agent Guards** – Rule-based safety checks for AI agent workflows, shipped as MCP servers, an API, a Claude plugin, and a GitHub Action. The test corpus, results, and misses are all published. github.com/mlawsonking/MCP
- **ForgeSense** – Founder. Vibration-sensing hardware on a physical test rig, with on-device signal processing, an edge service in Go, and a React dashboard.

SKILLS

Python · Frappe/ERPNext · REST APIs and systems integration · AI agent orchestration (Claude, Codex, MCP) · Docker-based deployment · Bash/Linux · Git/GitLab · CI/CD (GitLab CI, GitHub Actions) · SQL · Configuration management (CSA, ECR/ECO/ECN) · SolidWorks/CAD · Manufacturing QA

EDUCATION

B.S. Mechanical Engineering, The University of Texas at San Antonio, Dec 2022. *Summa cum laude*, 4.0 GPA.
B.S. Psychology, Texas A&M University, 2010.